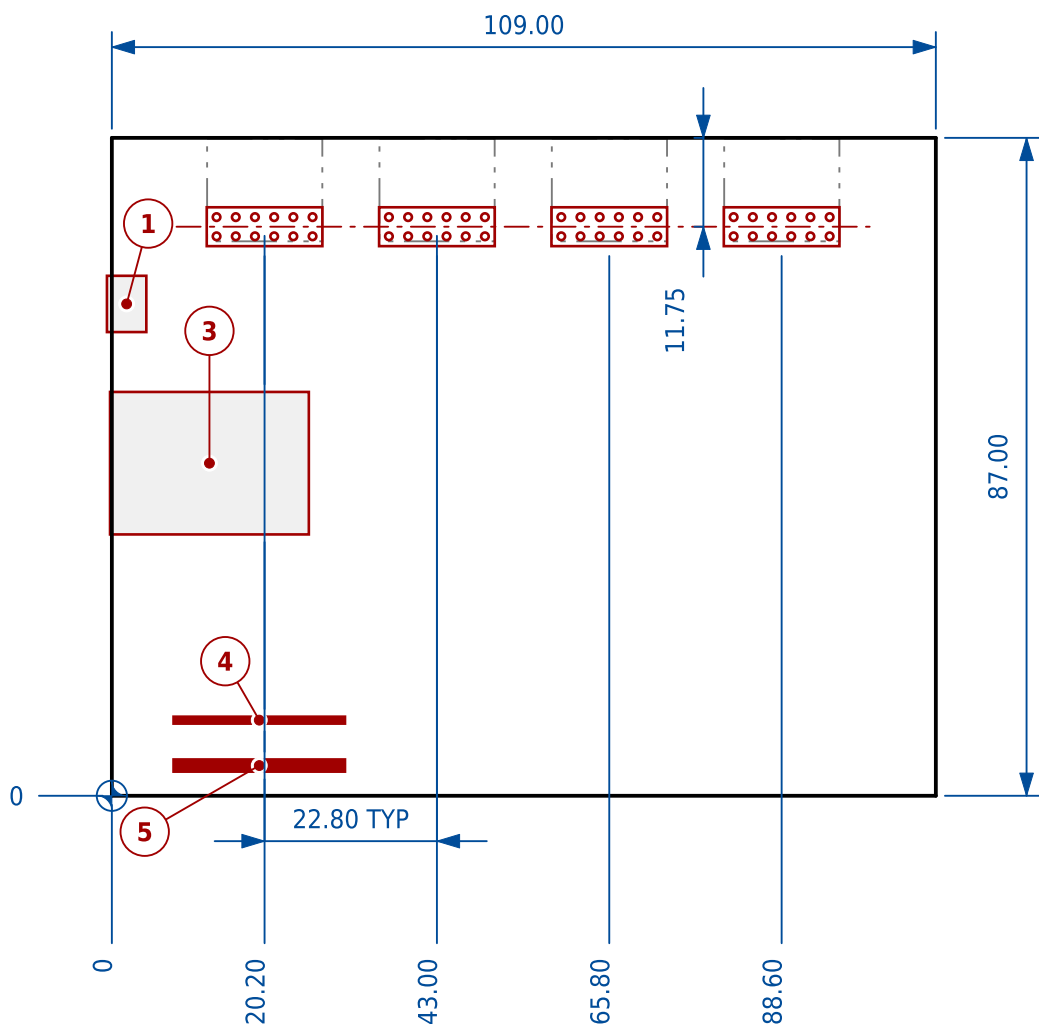




PMOD HOST HEADERS

| PORT | EDGE | CX mm  | CY mm  | PIN1 X mm | PIN1 Y mm |
|------|------|--------|--------|-----------|-----------|
| JA   | top  | 20.200 | 75.250 | 13.850    | 74.000    |
| JB   | top  | 43.000 | 75.250 | 36.650    | 74.000    |
| JC   | top  | 65.800 | 75.250 | 59.450    | 74.000    |
| JD   | top  | 88.600 | 75.250 | 82.250    | 74.000    |

FEATURE SCHEDULE

| # | FEATURE                                         | X EXTENT mm    | Y EXTENT mm    |
|---|-------------------------------------------------|----------------|----------------|
| 1 | micro-USB J10, programming and console          | -0.65 to 4.56  | 61.32 to 68.76 |
| 2 | Second USB port - not on this board             | -              | -              |
| 3 | Ethernet RJ45 J9                                | -0.24 to 26.06 | 34.56 to 53.40 |
| 4 | LEDs LD4-LD7, 4 on 7.00 mm pitch                | 8.16 to 30.84  | 9.54 to 10.46  |
| 5 | Tri-colour LEDs LD0-LD3, 4 on 7.00 mm pitch     | 8.16 to 30.84  | 3.19 to 4.80   |
| 6 | Expansion connector, first - not on this board  | -              | -              |
| 7 | Expansion connector, second - not on this board | -              | -              |
| 8 | Expansion connector, third - not on this board  | -              | -              |

LEGEND

- Board outline
- Component body, scheduled
- Pmod connector body
- Dimension, extension and leader

NOTES

- Viewed from the component side.
- No mounting holes. The drawing shows four 10.0 mm rubber feet centred 5.0 mm from each corner; a plate has to carry the board on those or clamp its edges.
- The Pmod pin rows are 2.50 mm apart on the drawing, not 2.54, and the hosts are on a 22.80 mm pitch where the PDF's dimension reads 0.90 in; both within the general tolerance.
- JA is nearest the USB and Ethernet corner, per the reference manual's callout figure.
- The 1.6 x 1.6 mm LED row nearest the edge is taken as the tri-colour LD0-LD3 and the 0603 row behind it as LD4-LD7; no source names them.
- Bodies read from the PDF plot are good to about +/-0.3 mm; the DXF figures carry the general tolerance.
- Assembled envelope, connector overhang included: 109.65 x 87.00 mm.

SOURCES

- Mechanical drawing: [https://digilent.com/reference/\\_media/reference/programmable-logic/arty-a7/arty\\_a7.zip](https://digilent.com/reference/_media/reference/programmable-logic/arty-a7/arty_a7.zip) - Digilent, Arty\_A7\_DXF.DXF and Mechanical\_Arty\_A7.pdf, dated 2020-09-02. Outline, pin fields, slots and pegs from the DXF; component bodies from the PDF, whose plot scale is recovered from the outline (axes agree to 0.20 %).
- 3D model: [https://digilent.com/reference/\\_media/reference/programmable-logic/arty/arty\\_rev\\_cad.zip](https://digilent.com/reference/_media/reference/programmable-logic/arty/arty_rev_cad.zip) - Digilent, Arty Rev C: used only to identify which of the PDF's outlines are the eight user LEDs.
- Reference manual: <https://digilent.com/reference/programmable-logic/arty-a7/reference-manual> - Pmod JA, JB, JC, JD; four tri-colour and four single LEDs.
- Mounting: <https://forum.digilent.com/topic/4523-410-319-mounting/> - Digilent staff, 2017-07-19: "There are no thru holes or mounting provisions"; the board stands on rubber feet.

DIGILENT ARTY A7

A7-35T and A7-100T

|                                              |                        |                                                                            |                    |                     |                 |
|----------------------------------------------|------------------------|----------------------------------------------------------------------------|--------------------|---------------------|-----------------|
| DRAWN<br><b>generated</b>                    |                        | VERSION<br><b>v0.0-186-gd944fb5</b>                                        | UNITS<br><b>mm</b> | SCALE<br><b>1:1</b> |                 |
| SIZE<br><b>A3</b>                            | SHEET<br><b>1 OF 1</b> | DRAWING NO<br><b>FPGA-ARTY-A7</b>                                          |                    |                     | REV<br><b>A</b> |
| MATERIAL<br><b>PCB, thickness not stated</b> |                        | GENERAL TOLERANCE<br><b>edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08</b> |                    |                     |                 |